ISS 2103 • EXAM REVISION PACK 00

Exam Map: Start Here

Built from the course outline and 5 past papers: 2018, 2023, SE2024, ISA2024, 2025

Priority: **READ FIRST**

Read this pack first. It shows what the examiner asks, how often, and in which pack each answer is. The other 8 packs each hold the past questions with model answers, short notes, diagrams, worked maths, and practice questions at three levels.

1 The exam format (same every year)

- **3 hours, 100 marks. FIVE** questions of 25 marks; answer **any FOUR**. Start each question on a fresh page.
- "Illustrate your answer, where appropriate, with large, clearly labelled diagrams" (ISA2024 instructions).
- One question **mixes topics**: e.g. ISA2024 Q1 = IP header + segmentation + LLC/MAC + subnetting. So you must cover **all** topics, not just four.

Time budget

$$\frac{180 \text{ min} - 10 \text{ min reading} - 10 \text{ min checking}}{4 \text{ questions}} = 40 \text{ min per question} \quad \leftarrow \textit{be strict}$$

$$\frac{40 \text{ min}}{25 \text{ marks}} = 1.6 \text{ min per mark} \quad \leftarrow \textit{a 6-mark part gets} \approx \textit{10 min}$$

Read all 5 questions first. Drop the one with the most parts you can't do. Do your **strongest** question first: it builds confidence and banks marks.

2 Your revision packs

Pack	Topic	Outline weeks	Priority
01	Fundamentals: components, transmission modes, LAN/MAN/WAN, topologies, client-server, protocols	1–2	HIGH
02	OSI & TCP/IP models, IP vs MAC, internetworking	1–2, 5–9	VERY HIGH
03	Physical layer: media, fibre, signals, impairments, modulation, multiplexing, switching, ADSL	1–2	VERY HIGH
04	Data link & MAC: LLC/MAC, CSMA/CD/CA, ALOHA, errors, flow control, bridges, switches, VLANs	3–4	VERY HIGH
05	Network layer: IPv4, **subnetting/VLSM**, IPv4 & IPv6 headers, SLAAC vs DHCPv6, DHCP, ARP	5–7	VERY HIGH
06	Routing (RIP, EIGRP, OSPF), congestion control, QoS, SDN, ISPs, SLA, WAN	5–7	HIGH
07	Transport & application: TCP vs UDP, handshake, DNS, troubleshooting, RPC, FTP, email	8–11	MEDIUM-HIGH
08	Security & emerging tech: threats, malware, DES vs AES, KDC, CA, VPN, 5G/IoT/cloud	8–9, 12	MEDIUM

3 Where the marks go: marks per topic per paper

Each cell shows the marks that topic earned in that paper (each paper sets 125 marks: 5 questions × 25). Darker means more marks.

	2018	2023	SE2024	ISA2024	2025	Total	ISS papers
01 Fundamentals	11	6	43	18	25	103	43
02 OSI & TCP/IP	12	4	13	15	26	70	41
03 Physical & switching	0	25	34	15	17	91	32
04 Data link & MAC	31	15	15	19	23	103	42
05 IP & subnetting	28	23	2	33	24	110	57
06 Routing & QoS	6	38	2	18	10	74	28
07 Transport & app	37	4	8	4	0	53	4
08 Security	0	10	8	3	0	21	3

Orange frame = the ISS (Information Security & Assurance) papers, the version you will sit. "ISS papers" = ISA2024 + 2025 marks.

EXAM TIP

Your paper is the **ISS 2103** (ISA) version. In the last two ISS papers, **IP/subnetting (57 marks)**, **fundamentals and topologies (43)**, **data link/MAC (42)** and **OSI (41)** dominate. Transport/application and security got almost nothing there, but don't skip them: they were big in 2018 and 2023.

4 The "bankers": most repeated questions

#	Question	Seen in	Pack
1	Subnetting / VLSM calculation	2018, 2023 (2×), ISA2024, 2025	05
2	OSI layers + functions (with diagram); OSI vs TCP/IP	all 5 papers	02
3	Network topologies: describe 3–4; ring vs bus	SE2024, ISA2024, 2025	01
4	Circuit vs packet (vs message) switching + examples	2023, SE2024, ISA2024, 2025	03
5	CSMA/CD (and CSMA/CD vs CSMA/CA)	2018, 2023, SE2024, ISA2024	04
6	TCP vs UDP; how UDP sends data; why TCP is reliable	2018, 2023, SE2024, ISA2024	07
7	IPv4 header fields (TTL, Protocol, Version, DS, addresses)	2023, ISA2024	05
8	IPv6: shorten addresses, advantages, header	2018, 2023, ISA2024	05
9	**SLAAC vs DHCPv6**	ISA2024, 2025 (the last two!)	05
10	LLC and MAC sub-layers	2023, ISA2024	04
11	Transmission media; fibre optic	2023, SE2024, 2025	03
12	Error detection (parity, CRC, checksum; single-bit vs burst)	2023, SE2024	04
13	Client-server vs peer-to-peer	SE2024, 2025	01
14	Routing: DV vs LS, RIP/RIPv2/EIGRP/OSPF	2023, ISA2024	06
15	Multiplexing (concept, FDM vs TDM)	SE2024, 2025	03
16	LAN/MAN/WAN; categories of networks	2018, ISA2024	01
17	Congestion control / network performance	ISA2024, 2025	06
18	Bridge / switch / router: function and layer	2018, SE2024, 2025	04

Likely "new" questions (in the outline but rarely asked)

- Sliding window: Go-Back-N vs Selective Repeat; IEEE 802 standards, FDDI, channel allocation (Pack 04).
- RPC; data compression; session-layer design issues; FTP, email (SMTP/POP3/IMAP), virtual terminals (Packs 02, 07).
- ISDN, terminal handling; backbone and local-access network design (Packs 03, 01).
- Emerging technologies: 5G, IoT, cloud, SDN (Packs 06, 08).

- Labs: `ipconfig`, `ping`, `tracert`, `netstat`, `nslookup`; straight-through vs crossover cables (Pack 07).

5 How to write answers that score

Command word	What the examiner wants
List / state / name	Short points only, one per mark.
Define / explain the term	A clear definition + one example (or how it works).
Describe / explain	Definition + how it works step by step + an example. A diagram earns marks.
Differentiate / distinguish / compare	A **table** with matching rows (4 marks = 4 rows). "Compare *and contrast*" also wants similarities.
With the aid of a diagram	The diagram must be **labelled**; usually about half the marks.
Calculate / show working	Every step: binary mask, borrowed bits, block size, then a results table.
Discuss	Several points, each explained, with advantages/disadvantages or examples.

TRAP – where students lose marks

- Writing one long paragraph for a "differentiate" question. Use a table.
- An unlabelled diagram. Label every box and arrow, and add a title.
- Giving 3 points for a 6-mark "explain" question. Go for 6 points, or 3 points each with a reason.
- Scenario questions (e.g. "HIT students accessing resources from home", "5-floor building"): answer **in the scenario's words**, not just theory.

6 Study plan: 22 Oct to 6 Nov

Days	Do
22–23 Oct	Pack 05: do every subnetting question in the pack **twice**, without looking.
24–25 Oct	Pack 02 (OSI): draw the OSI diagram from memory until it takes 3 minutes. Pack 01.
26–27 Oct	Pack 04 (data link): CSMA/CD flowchart, ALOHA, LLC/MAC, error table.
28–29 Oct	Pack 03 (physical, switching, multiplexing, ADSL).
30–31 Oct	Pack 06 (routing, congestion, QoS, SDN).
1 Nov	Packs 07 and 08.
2–3 Nov	Sit the **ISA2024** and **2025** papers under timed conditions (3 h each). Mark with the packs.
4–5 Nov	Redo every practice question you got wrong. Read all 9 cheat sheets.
6 Nov	Morning: this page + the Pack 05 cheat sheet. **Good luck!**

ONE-PAGE CHEAT SHEET

Models

- OSI 7→1: Application, Presentation, Session, Transport, Network, Data Link, Physical (*All People Seem To Need Data Processing*)
- PDUs: data / segment / packet / frame / bits
- Devices: L1 hub, repeater; L2 switch, bridge; L3 router
- Delivery: L2 hop-to-hop (MAC), L3 host-to-host (IP), L4 process-to-process (port)
- TCP/IP: Application, Transport, Internet, Network Access

Subnetting

- Hosts = $2^h - 2$; subnets = 2^s; block = 256− mask octet
- /24 254, /25 126, /26 62, /27 30, /28 14, /29 6, /30 2
- A 1–126 /8, B 128–191 /16, C 192–223 /24
- Private: 10/8, 172.16–31, 192.168/16

Headers

- IPv4 20 B; TTL; Protocol 6 TCP, 17 UDP, 1 ICMP
- IPv6 128-bit, 40 B, hop limit, flow label; `::` only once
- SLAAC = host builds its address (stateless); DHCPv6 = server assigns and records (stateful)
- DHCP DORA, UDP 67/68

Data link

- CSMA/CD wired: detect, jam, backoff. CSMA/CA wireless: avoid, RTS/CTS, ACK
- Pure ALOHA 18.4% (vulnerable $2T$); slotted 36.8% (T)
- Single-bit → parity; burst → CRC; checksum in IP/TCP/UDP
- Hamming: detect s needs $d = s + 1$; correct t needs $d = 2t + 1$
- MAC 48-bit = 24-bit OUI + 24-bit NIC. ARP: IP → MAC

Physical

- Circuit = dedicated path (phone, ISDN); packet = shared (IP, X.25, Frame Relay)
- FDM frequency (analogue); TDM time slots (digital); WDM light
- Guided: twisted pair, coax, fibre. Unguided: radio, microwave, infrared
- Impairments: attenuation, distortion, noise

Routing & transport

- RIP: DV, hop count (max 15), 30 s; v2 classless, multicast 224.0.0.9, auth
- EIGRP: advanced DV, DUAL, bandwidth + delay. OSPF: LS, Dijkstra, cost
- QoS: bandwidth, latency, jitter, loss. VoIP <150 ms, <30 ms jitter, <1% loss
- TCP: SYN, SYN/ACK, ACK. UDP: 8-byte header, no ACK. DNS: UDP 53
- DES 56-bit key, 64-bit block, 16 rounds; AES 128/192/256, 128-bit block